A Catholic Introduction to Artificial Intelligence Class 1 Lecture

Welcome to A Catholic Introduction to Artificial Intelligence Class 1!

"The question is not whether intelligent machines can have any emotions, but whether machines can be intelligent without any emotions." – Marvin Minsky

Learning Objectives

By the end of this class, students will be able to:

1. Understand the course objectives, requirements, and expectations.
2. Define Artificial Intelligence (AI) in their own words.
3. Distinguish between AI, Machine Learning (ML), and Deep Learning (DL).
4. Identify examples of AI in their daily lives.
5. Begin to think critically about what AI is and what it is not.

1A. Course Objectives

The Catholic Church views science to explore and comprehend the natural world created by God. Technology is encouraged when it respects ethical principles and upholds the dignity of human life. Catholics recognize the benefits of technological advancements for improving human life and society. Moral discernment is crucial to guide technology to serve humanity ethically and avoid harm or injustice.

This course integrates faith and technology to explore faith, ethics, and AI advancements together. This course will not teach students how to exploit AI concepts that violate academic integrity. Its main goal is to introduce the student AI concepts and the technology that makes it happen. Over the course of the topics, students will understand the so-called truth that AI provides and the truth that it doesn't provide. And it will cut through the fantasy and hype so often attributed to AI.

To summarize:

- Become critical thinkers and responsible users.
- Understand how AI impacts privacy, ethics, employment, and truth.

1B. Course Prerequisites

One of the following or equivalent:

- Introduction to Computer Science, Part 1 or Part 2
- Introduction to Computer Science Using SNAP!
- Computer Programming 101 with Python
- C# Programming with Unity 101

1C. Course Requirements

This course uses the following AI-based tools:

- `NotebookLM` is an AI research and study tool developed by Google that helps users understand and work with their own documents. Instead of answering questions from the general internet, it analyzes sources you upload—such as PDFs, Google Docs, or notes—and generates summaries, explanations, study guides, and even audio-style discussions based only on those materials. This makes it especially useful for research, learning, and organizing information because the AI stays grounded in the sources you provide. A Google account is required.
- `Jupyter Notebook` is an open-source interactive computing environment where users can write code, display results, add explanations, and create visualizations all in one document. It is widely used in data science, machine learning, and AI because it allows programmers to run code step-by-step while documenting their work. This makes it especially useful for learning, experimentation, and sharing reproducible research. Jupyter Notebooks can be created and execute within a browser with no need to install any additional software. No account registration required.
- `ChatGPT` (https://chatgpt.com/): an AI assistant that explains ideas, answers questions, researches information, and helps with writing, learning, planning, and problem-solving across many topics (from Chat GPT). No account registration required.
- `Google Gemini` (https://gemini.google.com/app): I am Gemini 3 Flash, an adaptive AI collaborator. I'm here to help you create text, images, music, and video with clarity and wit (from Gemini). A Google account is required.
- `Claude` (https://claude.ai/login): an AI assistant made by Anthropic. I can answer questions, write, analyze, code, and help with a wide range of tasks (from Claude). Account registration is required.
- `Google Teachable Machine` is a web tool from Google that lets users train simple machine-learning models—like image, sound, or pose recognition—without coding. A Google account is required.
- (Optional) Visual Studio Code (VSC): VSC is an integrated development environment (IDE) that aids in the writing, debugging, and execution of a large range of computer programs. It also can handle Jupyter Notebooks. Students with computer programming experience and who are familiar with VSC while find using VCS for

Jupyter Notebooks quite easy. VSC software is free but it must be installed on the student's computer or laptop.

- (Optional) Google Colab: (short for Google Colaboratory) is a free cloud-based environment from Google that lets users write and run Python code in a web browser. It works like an online notebook where you can mix code, text explanations, and visualizations—making it popular for data science, machine learning, and AI experiments. Because it runs in the cloud and can provide access to GPUs, students can build and test AI models without installing software on their own computers (from ChatGPT). A Google account is required.

1D. Course Expectations

This is a `live` course - you are expected to attend the live session each week. The recording is not intended to be viewed for the first time; it is intended as a refresher after you have attended the weekly live class.

Participation - use microphones. Chat for questions or raise your hand.

Essays

Academic Integrity -

1	NO AI	The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. **AI must not be used at any point during the {class | assignment.}**
2	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. **No AI content is allowed in the AI must not be used at any point during the {class | assignment.}.**
3	AI-ASSISTED EDITING	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. **AI can be used, but your original work with no AI content must be provided in an appendix.**
4	AI TASK COMPLETION, HUMAN EVALUATION	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. **You will use AI to complete specified tasks in your {assignment | assessment.} Any AI created content must be cited.**
5	FULL AI	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. **You may use AI throughout your {assignment | assessment.} to support your own work and do not have to specify which content is AI generated.**

1E. Assessment Overview

- Participation and discussions - 20%
- Short essays - 30%
- Quizzes- 10%
- Final Project - 40%

2. Define Artificial Intelligence (AI) in your own words.

What is AI to you? Define AI in your own words.

2A. What is Artificial Intelligence (AI)?

- Definition: AI is the field of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence. This includes problem-solving, understanding language, recognizing patterns, and making decisions.
- Examples: Siri, Google Translate, facial recognition, Netflix recommendations, email spam filters
- Misconception: AI is not conscious, sentient, or creative. Most current AI is "narrow" or task-specific.

2B: AI Capabilities

- `Narrow` AI that is good at one specific task (or a narrow set of tasks)
  - Examples are: spam filters, fraud detection models, recommendation systems, voice assistants, image classifiers, generative AI (ChatGPT, Copilot, Claude, etc.)
  - Each of these systems can only perform well in its own domain
  - Most AI systems today are `narrow` or `weak` AI
- `General` AI can learn and performan any intellectual taks a human can
  - Not a pre-programmed task
  - Examples are: learn a new job without training, reason across domains, transfer knowledge
- `Superintelligence` AI that is more capable that humans at essentially all intellectual tasks
  - Can reason, create, discover, strategize, and socially understand all on its own
  - Far beyond humans

3. Distinguish between AI, Machine Learning (ML), and Deep Learning (DL).

AI is the broad concept of machines doing intelligent things.

- Machine Learning (ML) is a subset of AI that uses data and algorithms to "learn" from experience using `labels`.
- Deep Learning (DL) is a further subset of ML that uses artificial neural networks to model complex patterns.

- Analogy: Think of AI as the universe, ML as a galaxy, and DL as a solar system within that galaxy.
- Supervised Learning is
- Unsupervised Learning is ...

4. Identify examples of AI in their daily lives.

Examples of AI in Everyday Life:

- Smartphones: Voice assistants, auto-correct.
- Streaming Services: Recommendation engines.
- Social Media: Feed algorithms, facial tagging.
- Navigation: Google Maps predicting traffic.
- Retail: Chatbots, personalized ads.

5. Ethics and Bias Teasers

- AI can make mistakes (e.g., false arrests due to facial recognition errors).
- Data used to train AI can be biased.
- Discussion: Should we always trust AI?
- Discussion: Who is responsible when AI is wrong?

6. Exit Ticket / Assessment:

"What is one way you've used AI without realizing it? What do you hope to learn in this course?"